Appendix C (Effect of Slope)

In Figure C1, *c* is varied from 0.04 to 0.18 in steps. Parts (a) and (b) represent the joint angles. Part (c) shows the profiles as a function of time. Part (d) represents the profiles as a function of distance from the stance leg. Part (e) shows the velocity of the horizontal moment, and Part (f) represents the velocity of the vertical movement for different values of *c*. During the time interval 0-1 s, leg 2 is the swinging leg and leg 1 is the stance leg, and during the time interval 1-2 s, it is vice versa.

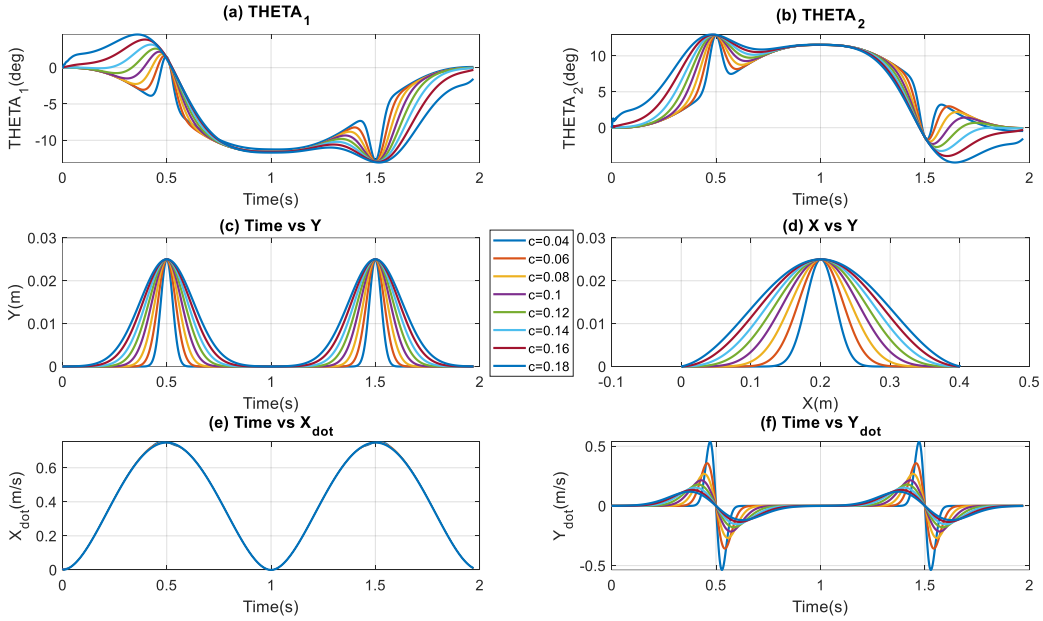


Figure C1: Parameters of the robot as a function of profile slope for a constant obstacle height

At too-low values of *c*, the slope of the profile is high, and more torque is required to move the swinging leg (Figure C2). Control energy is minimal when *c* is in the range of 0.12 -0.14 (Figure C3). However, if the obstacle is not at the center of the step length, the optimal value for *c* will differ.

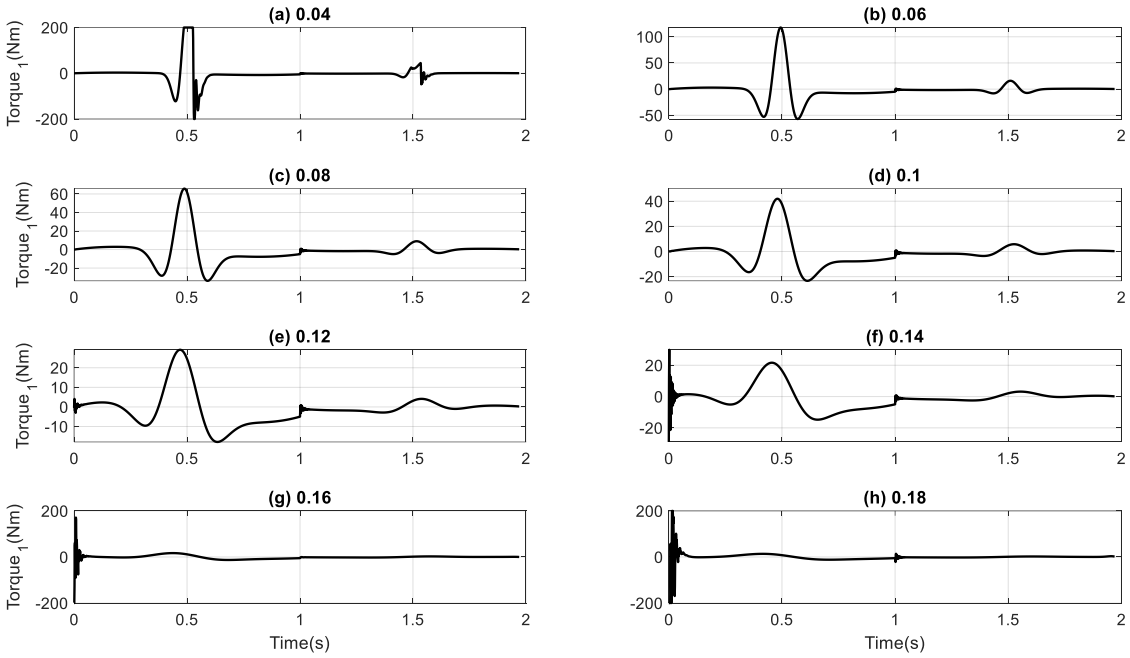


Figure C2: Torque at joint 1 as a function of profile slope for a constant obstacle height

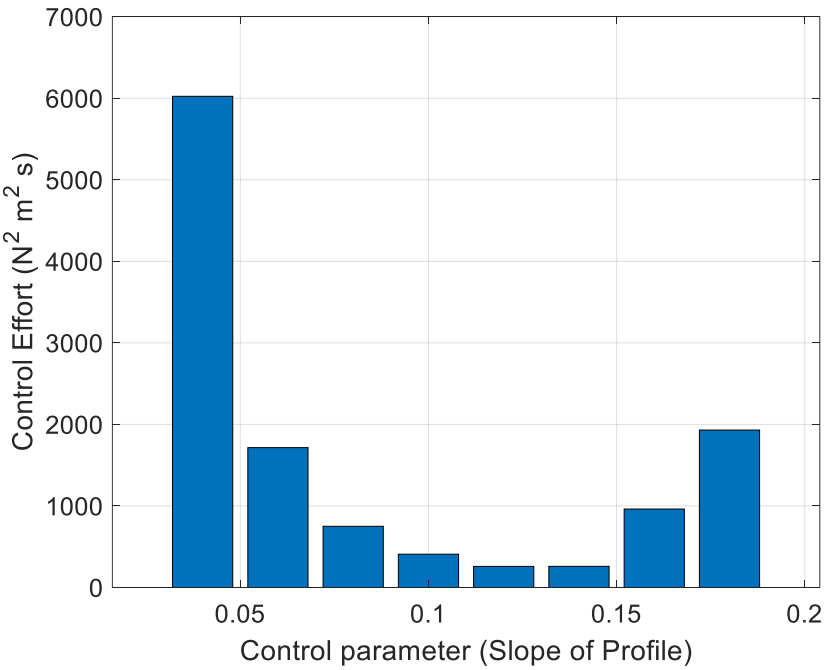


Figure C3: Control effort as a function of profile slope for a constant obstacle height

In Figure C4, the obstacle of zero width is assumed to be located 0.1 m from the start (stance leg). Part (a) shows the horizontal displacement of the swinging leg. Part (b) shows the obstacle's location as a function of time (the slopes of the profiles on the left and right of t=0.36 s). Part (c) shows the vertical displacement as a function of the horizontal displacement of the swinging leg. Part (d) shows the vertical velocity of the swinging leg for different profiles.

The control energy used is minimal when the slope of the left-hand side profile is 0.1, and that of the right-hand side is 0.2 (Figure C5). Much lower values of the slope (high value of c) are the source of impact forces. The left column of Figure C6 is drawn at c=0.12 and the right column at c=0.18. An impact is generated at 1 second (for the right column). During that period, there is an abrupt variation in joint velocities at this instant, which causes an abrupt change in joint torques. For this figure, the obstacle of zero width is lying at the middle of the foot length, as in Figure C1.

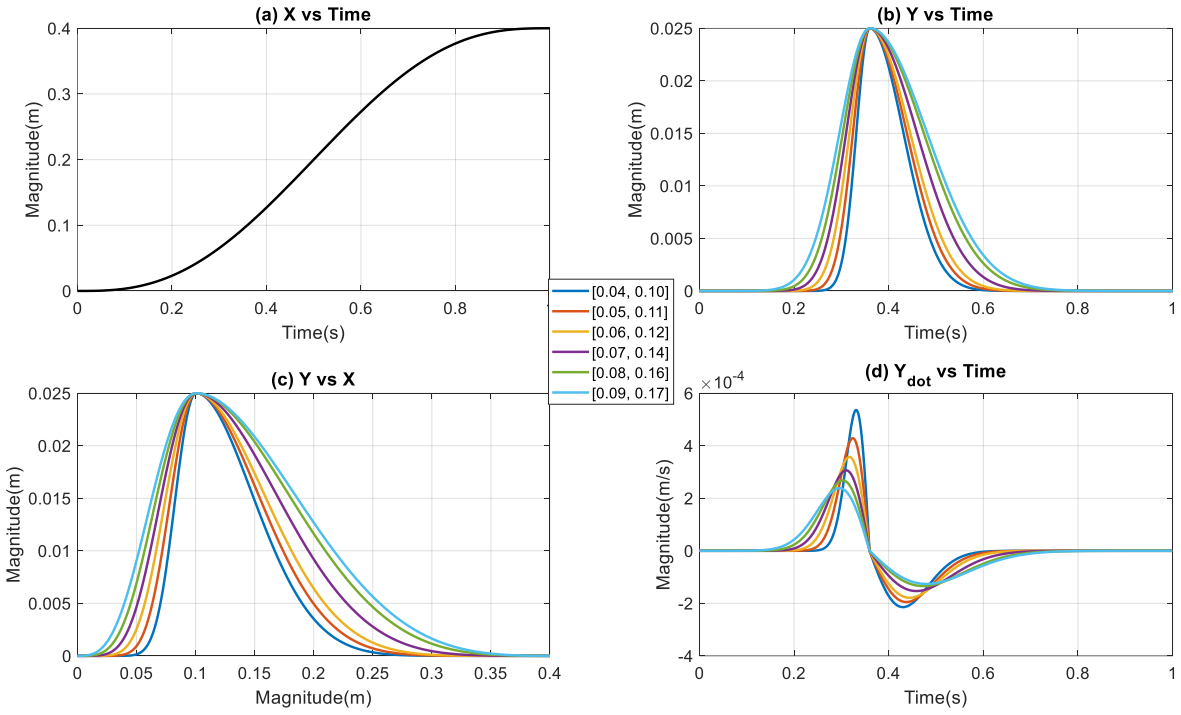


Figure C4: Parameters of the robot as a function of profile slope for a constant obstacle height (At a certain distance from the robot position)

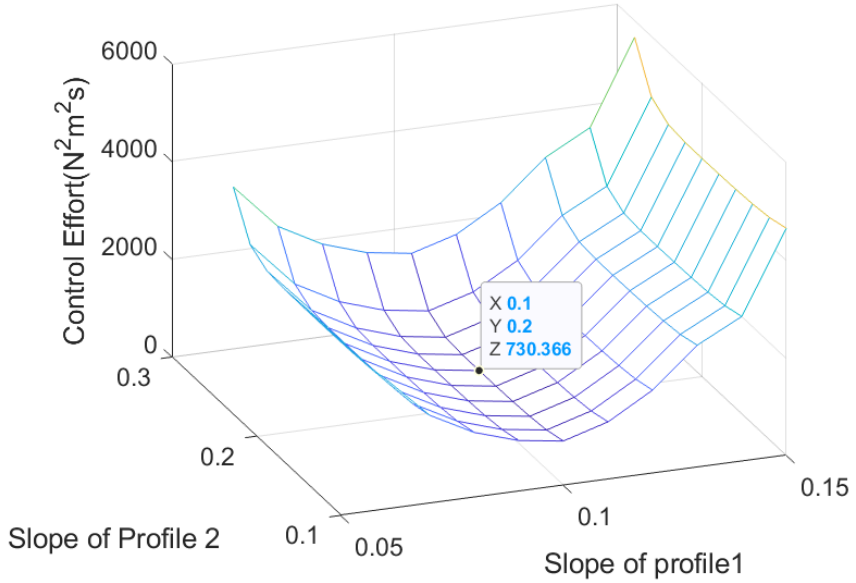


Figure C5: Control effort as a function of profile 1 and 2 slope for a constant obstacle height

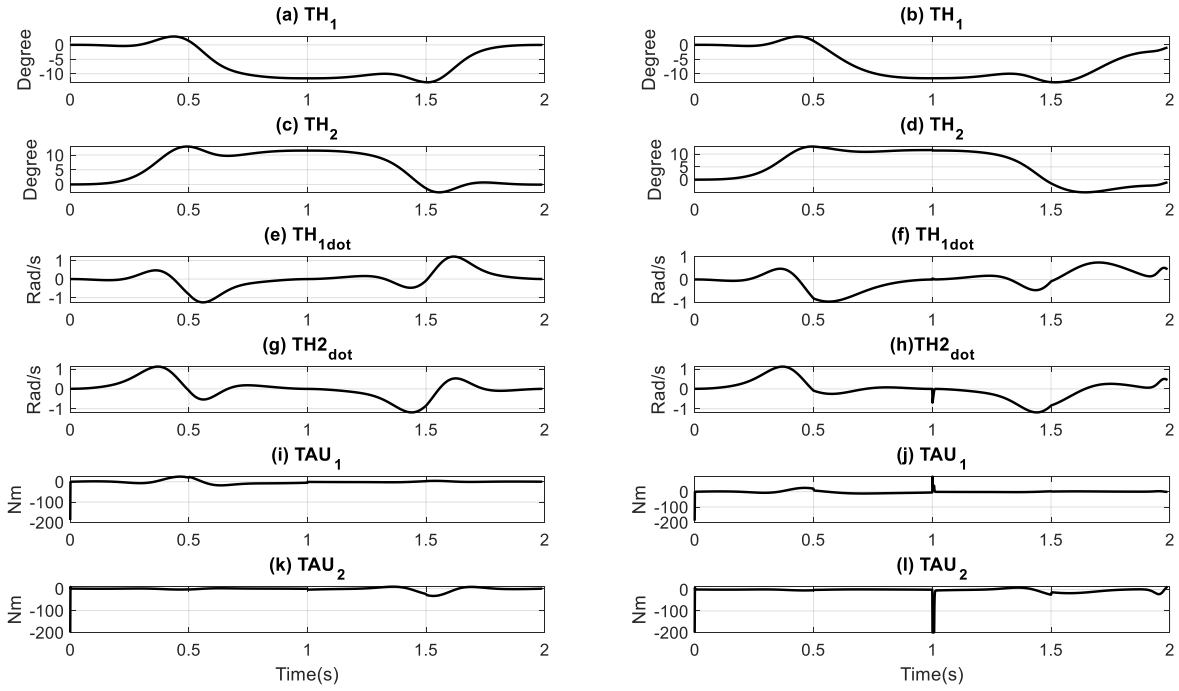


Figure C6: Effect of Impact forces